JOHANSSON ENGINEERING

# FLT Program — Regulatory Compliance Reference

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| **Document No.** | JDS-RPT-MEC-_____ |
| **Revision** | A |
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| **Status** | CURRENT |
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## 1. Purpose

This document is the **compliance proof** for the FLT program. It maps every program document (01-07) to the specific AFS 2017:3 sections it satisfies, including all three amendments. Use this document during audits, annual reviews, or when updating the program.

**Regulatory basis:** AFS 2017:3 consolidated with AFS 2019:1, AFS 2020:10, and AFS 2022:2.

## 2. Amendment Summary

All three amendments have been verified against the official PDF texts published by Arbetsmiljoverket.

## AFS 2019:1 (in force 2 September 2019)

| SECTION CHANGED | WHAT CHANGED | IMPACT ON FLT PROGRAM |
|---|---|---|
| 1 Kap. §2 | Scope clarified: vehicles excluded (EU directive refs added) | None — vehicles were never in scope |
| 1 Kap. §6 | Added definition: "enkla tryckkärl" (simple pressure vessels) | None — Doc 02 already classifies simple PVs |
| 3 Kap. §1, §5, §7 | CO2 fire extinguishers: max 10 years between inspections, specific int./ext. exam + pressure test | **Note:** If site has CO2 extinguishers, add to Doc 02 inventory |
| 5 Kap. §3 | First inspection must now include placement assessment (2 Kap §2-3) | **Covered:** Doc 02 section 6 includes placement risk |
| 5 Kap. §4 | When driftprov + int./ext. exam combined, driftprov done last (unless inspector says otherwise) | **Note:** Inform accredited body. No program doc change needed. |
| 6 Kap. §11 | Boiler monitoring: response time conditions changed from "1 AND 2" to "1 OR 2" | **Note:** Relevant only for boiler sites. Boilers meeting EITHER condition get extended response time. |

## AFS 2020:10 (in force 1 February 2021)

| SECTION CHANGED | WHAT CHANGED | IMPACT ON FLT PROGRAM |
|---|---|---|
| Transitional point 4 | Boiler operator certification: deadline extended to "5 years after entry into force" | **Superseded** by AFS 2022:2 |

## AFS 2022:2 (in force 30 May 2022)

| SECTION CHANGED | WHAT CHANGED | IMPACT ON FLT PROGRAM |
|---|---|---|
| Transitional point 4 | Boiler operator certification deadline: **1 December 2025** | **Action:** All boiler operators in category 1-4 must be certified by 1 Dec 2025. Verify in Doc 01 / Doc 04 personnel section. |

## Key Transitional Provisions (from base AFS 2017:3, as amended)

| POINT | REQUIREMENT | DEADLINE | STATUS |
|---|---|---|---|
| 3 | Lifetime journal (4 Kap §18) mandatory | 1 Dec 2019 | **Active** — Doc 06 |
| 4 | Boiler operator certification (6 Kap §4) | 1 Dec 2025 | **Past deadline** |
| 6 | First inspection not required for pre-2017 equipment | Permanent | Noted |
| 9 | Equipment newly in scope must have first recurring inspection | 1 Dec 2020 | **Past deadline** |

# 3. Program Document Compliance Map

## Doc 01 — Coordination Responsibility

| AFS SECTION | REQUIREMENT | HOW DOC 01 ADDRESSES IT |
|---|---|---|
| 4 Kap. §14 | Physical person must coordinate all work on Class A/B equipment | Section 2: named coordination person with contact details |
| 4 Kap. §14 | Written work instructions must exist | Section 4: responsibility checklist includes written instructions |
| 4 Kap. §14 | Must cover assembly, installation, maintenance, inspection | Section 4: all activities listed in responsibility checklist |
| 4 Kap. §17 | Physical person responsible for ensuring supervision is documented | Section 2: coordination person appointed and signed off |

## Doc 02 — Equipment Inventory

| AFS SECTION | REQUIREMENT | HOW DOC 02 ADDRESSES IT |
|---|---|---|
| 4 Kap. §15 | Register of all Class A and B equipment at each workplace | Sections 1-3: vessel identification, technical data, classification |
| 2 Kap. §6 pt 5 | Equipment correctly marked | Section 7: documentation checklist includes nameplate |
| 4 Kap. §10 | Classification per PS x V and fluid group | Section 3: classification table with rules |
| 4 Kap. §10 | Air/nitrogen Class B exemption | Section 3: classification rules note the exemption |
| 2 Kap. §2-3 | Placement requirements | Section 6: placement risk assessment per vessel |
| **AFS 2019:1** 5 Kap. §3 | First inspection includes placement assessment | Section 6: placement covered at inventory stage |

## Doc 03 — Risk Assessment

| AFS SECTION | REQUIREMENT | HOW DOC 03 ADDRESSES IT |
|---|---|---|
| 2 Kap. §1 | Employer must assess risks considering 5 factors | Section 2: risk assessment per vessel |
| 2 Kap. §1 pt 1 | Experience from use | Covered in risk identification |
| 2 Kap. §1 pt 2 | Remaining lifetime information | Covered; links to Doc 06 |
| 2 Kap. §1 pt 3 | Repairs and modifications | Covered in hazard identification |
| 2 Kap. §1 pt 4 | Accidents and incidents | Covered in hazard identification |
| 2 Kap. §1 pt 5 | Deviation reports and inspection results | Covered in hazard identification |

## Doc 04 — Supervision Program

| AFS SECTION | REQUIREMENT | HOW DOC 04 ADDRESSES IT |
|---|---|---|
| 4 Kap. §17 | Documented supervision routines for Class A/B | The entire document IS the documented routine |
| 4 Kap. §17 | Must show compliance with 2 Kap §6 points 1-6 | Section 1: all 6 mandatory checks listed with intervals |
| 4 Kap. §17 | Evaluate and revise at least once per year | Section 7: approval block with next review date |
| 4 Kap. §17 | Named physical person responsible | Section 5: personnel with named individuals |
| 2 Kap. §6 pt 1 | Equipment functions satisfactorily | Section 1, check 1 |
| 2 Kap. §6 pt 2 | No leaks have occurred | Section 1, check 2 |
| 2 Kap. §6 pt 3 | No harmful external or internal impact | Section 1, check 3 |
| 2 Kap. §6 pt 4 | No other faults or deviations | Section 1, check 4 |
| 2 Kap. §6 pt 5 | Correctly marked | Section 1, check 5 |
| 2 Kap. §6 pt 6 | Prescribed inspections carried out | Section 1, check 6 |
| 4 Kap. §16 | Continuous or periodic monitoring for Class A/B | Section 3: intervals defined per vessel and class |
| **Sanction** | Non-compliance: 10,000 — 100,000 SEK | Noted in document header |

## Doc 05 — Supervision Round Record

| AFS SECTION | REQUIREMENT | HOW DOC 05 ADDRESSES IT |
|---|---|---|
| 2 Kap. §6 | Six mandatory checks must be performed | Section 2: checks 1-6 map directly to §6 points 1-6 |
| 4 Kap. §17 | Supervision must be documented | The entire document IS the documentation |
| 4 Kap. §19 | Deviation reports for damage/deterioration | Section 3: findings with severity and action |
| 4 Kap. §19 pt 1-7 | Deviation report must contain 7 specific items | Section 3-4: description, severity, action, follow-up |

## Doc 06 — Lifetime Journal

| AFS SECTION | REQUIREMENT | HOW DOC 06 ADDRESSES IT |
|---|---|---|
| 4 Kap. §18 | Journal showing remaining lifetime for Class A/B | Sections 3-7: design data, measurements, calculations, summary |
| 4 Kap. §18 | Different parts described separately if different lifetimes | Section 4: per-vessel measurement records with location detail |
| 4 Kap. §18 | Lifetime extension analysis required if design life reached | Section 7: lifetime summary with Safe/Monitor/Critical/Expired status |
| 2 Kap. §1 pt 1-5 | Previous experience assessment | Section 1: 5 risk factors assessed per vessel |
| 2 Kap. §1 pt 2 | Remaining lifetime information | Sections 5-7: corrosion rate, fatigue, creep tracking |

### Doc 07 — Annual Review

| AFS SECTION | REQUIREMENT | HOW DOC 07 ADDRESSES IT |
|---|---|---|
| 4 Kap. §17 | Routines evaluated and revised at least once per year | The entire document IS the annual evaluation |
| 4 Kap. §17 | Must still show compliance with 2 Kap §6 | Section 1: round completion tracking |
| 4 Kap. §15 | Equipment register kept current | Section 4: equipment changes tracked |
| **AFS 2022:2** | Boiler operator certification by 1 Dec 2025 | Section 7: personnel competence status |

## 4. Inspection Intervals (Bilaga 1)

**Important:** Inspection intervals are NOT fixed. The accredited inspection body determines intervals based on equipment condition. The table below shows the **base intervals** (Bilaga 1 §1.4.1). Actual intervals may be shorter or longer.

## 4.1 Driftprov (Operational Test) — Base Intervals

| EQUIPMENT TYPE | BASE INTERVAL |
|---|---|
| Refrigeration/heat pump vessels | 4 years |
| Air, nitrogen, noble gas vessels (Group 2a) | 4 years |
| Vacuum-insulated condensed gas vessels | 4 years |
| LPG storage vessels | 4 years |
| Liquid-phase vessels (thermal expansion only) | 4 years |
| **All other pressure vessels** | **2 years** |
| Cisterns | 3 years |
| Piping connected to cisterns | 3 years |
| Other piping | 4 years |

**Extended:** If safety equipment functioned without action at 2 previous tests, max interval = **4 years** (§1.4.2).

**Shortened:** If safety equipment did NOT function, next interval is **halved** (§1.4.3).

## 4.2 Internal/External Examination — Condition-Based

The inspection body sets the interval based on condition. Progressive system:

| INTERVAL | KEY CONDITIONS |
|---|---|
| 4 years | Not fire-affected, remaining life > 5 yr, low crack risk, non-harsh environment |
| 6 years | Above + at least 1 clean 4-year inspection, no fatigue/creep |
| 8 years | Above + at least 2 clean inspections, no corrosion/erosion |
| 10 years | Above + at least 2 progressively longer inspections, no degradation |
| 2 years | 4-year conditions not met, but stable and > 3 yr before corrosion action |
| 1 year | 2-year conditions not met, but safe for 1 year per inspector |
| 6 months | 1-year conditions not met, but safe for 6 months |

## 4.3 Exemptions from Internal/External Examination (5 Kap. §6)

| EQUIPMENT | CONDITION |
|---|---|
| District heating heat exchangers | Temperature max 120 C |
| Vacuum-insulated CO2 vessels | — |
| Deep-cooled condensed gas vessels | Boiling point below -153 C |
| Hydraulic oil system vessels | Low crack risk material |
| Refrigeration/heat pump vessels | Non-corrosive medium |
| Water piping | Max 120 C, protected installation |

## 5. Sanctions

| VIOLATION | FINE |
|---|---|
| No documented supervision routines (4 Kap §17) | 10,000 — 100,000 SEK |
| Pressurising without valid inspection (5 Kap §1) | 15,000 — 300,000 SEK |
| Operating boiler without valid monitoring (6 Kap §1-2) | 15,000 — 300,000 SEK |

## 6. Annual Compliance Checklist

Use this checklist during the annual review (Doc 07) to verify the program remains compliant:

- [ ] **4 Kap §14:** Coordination person appointed and documented (Doc 01)
- [ ] **4 Kap §15:** Equipment register up to date (Doc 02)
- [ ] **4 Kap §16:** Monitoring type (continuous/periodic) documented per vessel (Doc 04)
- [ ] **4 Kap §17:** Supervision routines documented and reviewed this year (Doc 04, Doc 07)
- [ ] **4 Kap §17:** Named person responsible for supervision (Doc 01, Doc 04)
- [ ] **2 Kap §6:** All 6 mandatory checks addressed in every round (Doc 05)
- [ ] **4 Kap §18:** Lifetime journal maintained for limited-life equipment (Doc 06)
- [ ] **4 Kap §19:** Deviation reports completed for all damage/deterioration (Doc 05 findings)
- [ ] **2 Kap §1:** Risk assessment current and considers all 5 factors (Doc 03)
- [ ] **2 Kap §2-3:** Placement assessment completed for all vessels (Doc 02)
- [ ] **5 Kap §1:** All formal inspections current — no expired certificates
- [ ] **6 Kap §4:** Boiler operators certified (if applicable) — deadline was 1 Dec 2025
- [ ] **AFS 2019:1:** CO2 fire extinguishers included in inventory (if present on site)
- [ ] **AFS 2019:1:** First inspection includes placement assessment (5 Kap §3)

# Revision History

| REV | DATE | AUTHOR | DESCRIPTION |
|---|---|---|---|
| A | 2026-04-11 | N. Johansson | Initial release — verified against official PDF texts of AFS 2017:3, AFS 2019:1, AFS 2020:10, AFS 2022:2 |